

On Representation

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(Content developed with R. Mizoguchi)

Summary

- Representation



- A representation can be:**
- a material object
 - an information obj
 - an event





Information Obj (?)



Material Obj (?)

**Are representations
spatio-temporal entities?**

Event (?)



From Information to Representation

- Let's be more precise, by representation we mean **something** that says **something** about **something**

From Information to Representation (view IR)

- Let's be more precise, by representation we mean **something_1** that says **something_2** about **something_3**
- If you want to impress people:
“a representation is a **content-bearing** entity...”
- Usually this notion is specified in terms of **information**:
what represents (**something_1**) and what is represented (**something_3**) are connected by an information entity (**something_2**)
e.g., a common structure, a similar pattern, a pragmatic association
- In this view, information is an entity which is a constituent of representation:
You need to understand information to understand representation

Quick check

- Is there a representation in this room?
- Is there something which **is not** a representation in this room?
- Representation is a broad notion but not too broad:
 - (I) Irreflexive
 $\neg \text{Representation}(x,x)$
 - (II) Not symmetric
 $\text{Representation}(x,y) \rightarrow \neg \text{Representation}(y,x)$
 - (III) Not transitive (sorites paradox)
 $\exists x,y,z \text{ Representation}(x,y) \wedge \text{Representation}(y,z) \wedge \neg \text{Representation}(x,z)$

Back to information

- **Data, information and knowledge** are highly debated notions
Nonetheless, it is common to assume them as **ontologically different**
- Yet, any discussion about data, information, or knowledge is also a discussion of **representations used in different manners**
- This suggests that the distinction across data, information, and knowledge might be **contextual**
- Indeed, one can differentiate data, information, and knowledge based on how a representation is used:
 - **data** is a representation when processed
(a product code stored in a repository)
 - a piece of **information** is a representation when exchanged
(a product code sent to another repository) —no need to have a purpose
 - a piece of **knowledge** is a representation when acquired/learned
(a product code received by a repository)

From Representation to Information (view RI)

- The long standing problems with the IR view leads to study the alternative RI
- Essentially, we reverse our assumption on dependence:
instead of studying information to understand representation,
we study representation to understand information, thus in RI

Representation is more fundamental than information

- What do we need to address to talk about representation in view RI?

Summary (updated)

- The **form-content relationship** in representation: OReO
- At the core of representation with **music** and **drama**
- Happily leaving **aboutness** out

Material from:

- ***“An Ontology of Representation”***, R. Mizoguchi and S. Borgo, *Applied Ontology* Vol. 20(2) 137–152, 2025 DOI: 10.1177/15705838251337943
- Entries on “representation” from the *Stanford Encyclopedia of Philosophy*
- Entry “Scientific representation” from the *Internet Encyclopaedia of Philosophy*

Informal example: music

In terms of representation, what is a “piece of music”?

- (M1) The musical artwork as conceived by the composer (a mental object)
- (M2) The musical score (a text with associated semantics)
- (M3) What the composer coded in the music score (sound sequence specification)
- (M4) The performance of the player (an activity)
- (M5) The sound generated by the performance (an event)

Representing music

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OReO

- We define an **Ontological Representation Object (OReO)** to be a temporally localised object that is necessarily composed of two interrelated parts:
A **FORM**, i.e., an expression in the language (such expression could be realized e.g. in writing, but the expression is not its realisation as a material entity)
A **CONTENT** of that expression according to the language's semantics
Their relationship is called **ENCODING** (encoding≠encryption)
- It follows that “expressions” that do not have content according to the semantics for that language are not representations (e.g., some types of artwork)
- An OReO exists in time but not in space, as it is not a concrete entity (not even in a parasitic sense such as holes), and is independent of the existence of an agent that can understand or even recognize the object.
- (Here we do not discuss the relationship btw OReO and how it is used or created)

Representation v.s Representation-of

- The **OReO theory** is not to establish whether an OREO is a reliable representation or not, nor to establish whether there exists something in the world to which it refers
- In other words, aboutness is not part of representation and should be studied separately
Having an encoding of form and content is independent of what they might be about (if anything)
- Example: a procedure (a set of actions' descriptions in some, perhaps partial, order) is an OREO since it has a form and a content.
The form is an expression (not what you read in a paper or see in a picture),
the content is a specification (set of constraints) of actions and their execution ordering.
- An OReO-procedure is the encoding btw form and content; not the performed actions, nor the process that realizes it, nor the booklet that reports it.
- One can give a procedure that is physically or even logically impossible to realize:
it is still a representation as long as it has form, content and the encoding.

The core of the OReO (RI view)

- Representations are pairs composed of two concepts, a form and a content $\langle \text{form}, \text{content} \rangle$
- Form and content are in distinct concept subcategories depending on how they can be (or cannot be) realized
- Representation $\langle \text{form}, \text{content} \rangle$ satisfies a relationship called “encoding” (not investigated yet) relatively to some method, which is also a concept the nested
- Some forms are themselves representations, the pair pattern can be iterated:
 $\langle \langle \langle \text{form}', \text{content}' \rangle, \text{content} \rangle$
- The nested content' is needed to dealing with entities such as the style of writing or of playing
- This apparently simple and minimal machinery (all notions are concepts and relations across concepts) sets the basis to clarify and model the core of a theory of representation
- How far this approach can go remains to be investigated

Traditional view vs. OReO

- In the traditional view by representation one means

something_1 that says **something_2** about **something_3**

Representation

(no distinction btw representation and representation-of)

- In OReO one distinguishes representation and representation-of

something_1 that says **something_2** about **something_3**

OReO representation

OReO representation-of

Prioritising form or content

- Some OREOs focus primarily on content, others on form
- A procedure (the specification of a set of actions and their order of execution) can be expressed in different languages but its main raison d'être is the content that is independent of the used language
- Similarly for a drama (a specification of happenings, including dialog)
- In comparison, a piece of calligraphy focus on form
- This leads to a finer distinction of some types of representation:
A map as a specification (CAD file of an industrial product)
A map as a description (subway map)
- Q.: The portrait of a person gives priority to form or content?

Form and content realization

- There are two types of realizations (mapping from a source to a concrete entity):
form-realization takes as source the form of the representation
content-realization takes as source the content of the representation
- form-realization exists for all kinds of representation and consists in generating a physical entity where the form (e.g., a linguistic expression) is physically manifested (e.g., an ink pattern on a material support)
- usually form-realization admits multiple realization types some of which are object realizations (e.g., a sentence in a book; a map printed on paper) and others process realizations (e.g., uttering a sentence)
- content-realization exists for procedures, music and dramas (those for which content is a specification)

Content realizability

- Procedure, music, drama, map, novel, poem, handwriting, painting...
- Some OReOs have **realisable** content, they allow for performers (procedure, music, drama)
Other OReOs have **unrealisable** content (map, novel, poem, handwriting, and painting)
- What about reading a poem in front of an audience?
- A.: in this view, reading a poem is not a realisation of the content but of the form
- Painting vs. (piece of) music:
A musician is performing according to a content (specification)
A painter is creating something anew in terms of content
It follows that playing music and painting are distinct representations from the ontological viewpoint
- NOTE: performing can be creative but we distinguish *content-based performing* (following a specification) and *form-based performing*. In the latter, creativity applies to the style.

Examples

Drama and Novel (as representations)

- Both drama and novel are shared as pieces of written texts but:
The content of a drama is a specification characterizing events and is realized by actors
The content of a novel is not a specification
- Readers of a novel may read descriptions of events but do not realize a content
Reading a novel is comparable to attending (as a spectator) a drama performance, listening to a sequence of (musical) sounds, and watching a painting (the material object)
- None of these activities are performances, they are **perceptual experiences**

Examples

Piece of music (as a representation)

- A composer composes expressions (form) associated with the specification of sound sequences (content)
- There can be various performances of a piece of music and any two performances differ, all of them satisfy the specification coded in the musical score (with some tolerance, of course)
- What one listens to is a content-realization (a concrete instance of musical sound)
- A drama is for activities what a piece of music is for sounds

Examples

Procedure (as representation)

- The procedure (say, the quick sort algorithm) has a form (how it is presented in a suitable language) and such form may be realized on a sheet of paper (the writing of such expression, form-realisation)
- The form of the procedure depends on the chosen language
the content of the procedure (the specification) is language independent
- What the execution of a procedure (e.g., the quick sort algorithm) creates is an instance of a sorting activity, i.e., a content-realization of the algorithm
- Procedures and dramas are similar from the representational viewpoint

Examples

Map (as representation) [in cartography]

- Models of products (form-realization as CAD files) and maps (e.g., form-realization as city maps and subway maps) are different representations
- A map does not provide a specification, it does not prescribe what or how to produce/process something. The content of a map is a set of statements related to the spatial layout of some types of objects and covers properties and relationships about object distribution, matter configuration, distance proportion, and so on
- In more general terms, the content of a map is a description of objects with some of their properties and relationships (primarily spatial)
- As a representation, a map is for space what a novel is for events

IAO and OReO

- Information Artefact Ontology (Ceusters, 2012; Smith & Ceusters, 2015) has been influential in applied ontology
- IAO and OReO are clearly related but focus on different entities as the first gives priority to information, the latter to representation (cf. slide 16)
- IAO takes information to be an object, in OReO information is a role played by a representation in the context of an activity
- The notion of information content entity (ICE), the main type modeled in IAO, from OReO's viewpoint is ambiguous as it combines both content and form
- Information quality entity (IQE), a concretization of ICE, is similar to OReO's notion of Representing-thing (but this is not really clear for lack of discussion/examples in the main sources listed above)

A few points to remember

- In applied ontology **representation** is a core notion but is still not properly investigated
- There are important consequences in taking representation as more fundamental than information (**RI view** vs. **IR view**): are there other approaches to consider?
- In OReO, representation is a pair of **form** and **content** tied by an **encoding** relationship and an encoding method.
Form can always be realized (form-realization), only some types of OReO admit content-realization
- In OReO, **data**, **information** and **knowledge** are roles of a representation
- Content can be a **specification**, a **description** or a **denotation**
- In OReO, distinguishes representation and representation-of
- **Aboutness** is needed when a representation is used to refer to real world in a certain context
- A **representation** is a 'network of concepts'; a **representing-thing** is a form-realization

thanks